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Carrera Lic. En Ciberdefensa

Análisis Matemático I

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Ej: SEA

$$f(x)=\begin{cases}\left[1+2\operatorname{sen}^2(5x-5)\right]^{\frac{x-2}{(x-1)^2}} & \text{si } x<1\\ -\frac{1}{12} & \text{si } x=1\\ \dfrac{\sqrt{7+2x}-3}{x^2-6x+5} & \text{si } x>1\end{cases}$$

CALCULAR $\lim_{x\to 1} f(x)$ Y DETERMINAR SI f ES CONTINUA EN $x=1$

S/

$$\lim_{x\to 1^+} f(x)=\lim_{x\to 1^+}\frac{\sqrt{7+2x}-3}{x^2-6x+5}\ \overset{\text{(INDET. }\frac{0}{0}\text{)}}{=}\lim_{x\to 1^+}\frac{\sqrt{7+2x}-3}{x^2-6x+5}\cdot\frac{\sqrt{7+2x}+3}{\sqrt{7+2x}+3}=$$

$$=\lim_{x\to 1^+}\frac{(\sqrt{7+2x})^2-3^2}{(x^2-6x+5)(\sqrt{7+2x}+3)}=\lim_{x\to 1^+}\frac{7+2x-9}{(x^2-6x+5)(\sqrt{7+2x}+3)}=$$

$$=\lim_{x\to 1^+}\frac{2x-2}{(x^2-6x+5)(\sqrt{7+2x}+3)}=\lim_{x\to 1^+}\frac{2(x-1)}{(x-1)(x-5)(\sqrt{7+2x}+3)}=$$

$$=\lim_{x\to 1^+}\frac{2}{(x-5)\cdot(\sqrt{7+2x}+3)}=\frac{2}{-4\cdot 6}=-\frac{1}{12}$$

$$\lim_{x\to 1^-} f(x)=\lim_{x\to 1^-}\left[1+2\operatorname{sen}^2(5x-5)\right]^{\frac{x-2}{(x-1)^2}}=$$

(INDET. 1^∞)

(USAMOS: $\lim_{y\to 0}(1+y)^{\frac{1}{y}}=e$)

$$=\lim_{x\to 1^-}\left[\left(1+2\operatorname{sen}^2(5x-5)\right)^{\frac{1}{2\operatorname{sen}^2(5x-5)}}\right]^{2\operatorname{sen}^2(5x-5)\cdot\frac{x-2}{(x-1)^2}}$$

$y\to 0$ cuando $x\to 1^-$; el corchete interior $\to e$; el exponente $\to -50$

$$=e^{-50}$$

C.A.:

$$\lim_{x\to 1^-} 2\ln^2(5x-5)\cdot\frac{x-2}{(x-1)^2} =$$

$$= \lim_{x\to 1^-} 2\cdot\underbrace{\frac{\ln(5(x-1))}{5(x-1)}}_{\to 1}\cdot\underbrace{\frac{\ln(5(x-1))}{5(x-1)}}_{\to 1}\cdot\underbrace{(x-2)}_{\to -1}\cdot 25 = -50$$

(USAMOS $\lim_{t\to 0}\frac{\ln(t+1)}{t}=1$).

OBS.: $\lim_{x\to 1^-} f(x) = e^{-50} \neq -\frac{1}{12} = \lim_{x\to 1^+} f(x) \Rightarrow \nexists \lim_{x\to 1} f(x)$

$\therefore$ f NO ES CONTINUA EN $x=1$

Ej.: DETERMINAR EL DOMINIO DE $f(x) = 2+\ln(16-x^2)$.
HALLAR TODOS LOS x TALES QUE $f(x)=3$.

S/ $x\in Dom(f) \iff 16-x^2>0 \iff x^2<16 \iff \sqrt{x^2}<\sqrt{16}$

$\iff |x|<4 \iff -4<x<4$

$\therefore \boxed{Dom(f)=(-4,4)}$

$f(x)=3 \iff 2+\ln(16-x^2)=3 \iff \ln(16-x^2)=1 \iff$

$\iff 16-x^2=e \iff x^2=16-e \iff \sqrt{x^2}=\sqrt{16-e}$

$\iff |x|=\sqrt{16-e} \iff (\boxed{x=\sqrt{16-e}} \text{ o } \boxed{x=-\sqrt{16-e}})$.

$|x| = \sqrt{16-e} < \sqrt{16} = 4 \Rightarrow |x|<4 \Rightarrow \boxed{-4<x<4}$

$\therefore$ AMBAS SOLUCIONES ESTÁN EN $Dom(f)$. ✓

Ej: HALLAR a y $b \in \mathbb{R}$ TALES QUE LA SIGUIENTE FUNCIÓN SEA CONTINUA EN $x = 1$.

$$f(x) = \begin{cases} \dfrac{\operatorname{sen}(a(x-1))}{x^2+3x-4} & \text{SI } x > 1 \\ b & \text{SI } x = 1 \\ (5+2x^2-6x)^{\frac{1}{x-1}} & \text{SI } x < 1 \end{cases}$$

S/ $f(1) = b$

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} \left(5+2x^2-6x\right)^{\frac{1}{x-1}} = \lim_{x\to 1^-} \left(1+\underbrace{2x^2-6x+4}_{\to 0}\right)^{\frac{1}{x-1}} =$$

($5+2x^2-6x \to 1$; $\frac{1}{x-1} \to -\infty$)

$x < 1 \Rightarrow x - 1 < 0$

$$= \lim_{x\to 1^-} \left[\left(1+\underbrace{2x^2-6x+4}_{\to 0}\right)^{\frac{1}{2x^2-6x+4}}\right]^{(2x^2-6x+4)\cdot\frac{1}{x-1}} = e^{-2}$$

($\frac{1}{2x^2-6x+4} \to \frac{1}{y}$; $\left(1+\underbrace{2x^2-6x+4}_{\to 0}\right)^{\frac{1}{2x^2-6x+4}} \to e$; $(2x^2-6x+4)\cdot\frac{1}{x-1} \to -2$)

C.A.: $\lim_{x\to 1^-} \dfrac{2x^2-6x+4}{x-1} = \lim_{x\to 1^-} \dfrac{2\cancel{(x-1)}\cdot(x-2)}{\cancel{x-1}} = -2$ ($\cancel{x-1} \to 1$; $(x-2) \to -1$)

$$\therefore \lim_{x\to 1^-} f(x) = e^{-2}$$

$$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} \frac{\operatorname{sen}(a(x-1))}{x^2+3x-4} = \lim_{x\to 1^+} \frac{\operatorname{sen}(a(x-1))}{(x-1)\cdot(x+4)} =$$

(INDET. $\frac{0}{0}$)

$$= \lim_{x\to 1^+} \underbrace{\frac{\operatorname{sen}(\overbrace{a(x-1)}^{\to 0})}{\underbrace{a(x-1)}_{\to 0}}}_{\to 1} \cdot \frac{a}{x+4} = \frac{a}{5}$$

↖ DEBE SER $a = 0$, SINO, $\lim_{x\to 1^+} f(x) = 0 \neq e^{-2}$ Y f NO RESULTA CONTINUA EN $x = 1$.

$$\therefore \lim_{x\to 1^+} f(x) = \frac{a}{5}$$

como $f(1) = b$, $\lim_{x \to 1^+} f(x) = \frac{a}{5}$ y $\lim_{x \to 1^-} f(x) = e^{-2}$,

f es continua en $x = 1$ si y sólo si

$$b = \frac{a}{5} = e^{-2} \iff \boxed{a = 5.e^{-2} \text{ y } b = e^{-2}}$$

③

Ej.: Analizar si la siguiente función es derivable en $x = 0$. Si es posible, hallar la recta tangente al gráfico de f en $(0, f(0))$.

$$f(x) = \begin{cases} \dfrac{\ln(x^2+3) - \ln(3)}{x} & \text{si } x \neq 0 \\ 0 & \text{si } x = 0 \end{cases}$$

Si

$$\lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \to 0} \frac{\dfrac{\ln(x^2+3) - \ln(3)}{x}}{x} =$$

$$= \lim_{x \to 0} \frac{\ln(x^2+3) - \ln(3)}{x^2} \quad (\to 0 \text{ y } x^2 \to 0) = \lim_{x \to 0} \frac{\ln\left(\frac{x^2+3}{3}\right)}{x^2} =$$

$$= \lim_{x \to 0} \frac{1}{x^2} \cdot \ln\left(1 + \frac{x^2}{3}\right) = \lim_{x \to 0} \ln\left(\left(1 + \frac{x^2}{3}\right)^{\frac{1}{x^2}}\right) =$$

(indet $\infty \cdot 0$)

$$= \lim_{x \to 0} \ln\left(\left[\left(1 + \frac{x^2}{3}\right)^{\frac{3}{x^2}}\right]^{\frac{x^2}{3} \cdot \frac{1}{x^2}}\right) = \ln\left(e^{\frac{1}{3}}\right) = \frac{1}{3}$$

($\frac{3}{x^2} \to \infty$, $y \to 0$, $\frac{1}{y} \to \infty$, $\left(1 + \frac{x^2}{3}\right)^{\frac{1}{y}} \to e$)

$\therefore$ f es derivable en $x = 0$.

La recta tangente al gráfico de f en $(0, f(0)) = (0, 0)$ tiene ecuación

$y = f'(0)\cdot(x-0) + f(0)$ $\quad f'(0) = \frac{1}{3}$, $f(0) = 0$

$y = \frac{1}{3}\cdot(x-0) + 0$

$\boxed{y = \frac{1}{3}x}$

$y = \frac{1}{3}x + b$

$0 = \frac{1}{3}\cdot 0 + b$

$0 = b$

OBS: LA RECTA NORMAL EN $(0, f(0))$ TIENE ECUACIÓN $y = \frac{-1}{f'(0)}\cdot(x-0) + f(0)$ $\quad f'(0) = \frac{1}{3}$, $f(0) = 0$

$\boxed{y = -3x}$

$y = -3\cdot x + b_\perp$

$0 = -3\cdot 0 + b_\perp$

$0 = b_\perp$

EJ.: HALLAR LA ECUACIÓN DE LA RECTA TANGENTE AL GRÁFICO DE $f(x) = \frac{3x}{x+2}$ EN EL PUNTO $(4,2)$.

S/ $f(4) = 2$ $\qquad y = f'(4)\cdot(x-4) + f(4)$ $\quad f(4) = 2$

$f'(x) = \frac{3\cdot(x+2) - 3x}{(x+2)^2} = \frac{3x+6-3x}{(x+2)^2} = \frac{6}{(x+2)^2} = \frac{6}{6^2} = \frac{1}{6}$

$\therefore \boxed{y = \frac{1}{6}(x-4) + 2}$

$y = \frac{1}{6}x - \frac{4}{6} + 2$

$y = \frac{1}{6}x - \frac{2}{3} + 2$

$\boxed{y = \frac{1}{6}x + \frac{4}{3}}$ → RECTA TANGENTE AL GRÁFICO DE f EN $(4,2)$.

$y = \frac{1}{6}\cdot x + b$

$2 = \frac{1}{6}\cdot 4 + b$

$2 = \frac{2}{3} + b$

$\frac{4}{3} = b$

OBS: RECTA NORMAL: $y = -6(x-4)+2$

$y = -6x + b_1$

$2 = -6 \cdot 4 + b_1$

$26 = b_1$

$\boxed{y = -6x + 26}$

EJ: SEA f UNA FUNCIÓN DERIVABLE TAL QUE $f(0) = -2$ Y $f'(0) = 9$. CALCULAR $g'(0)$ SI

$$g(x) = 3\operatorname{sen}(x) \cdot f(x) + f(x^2) + f(\operatorname{sen}(x)).$$

S/

$$g'(x) = 3\cos(x) \cdot f(x) + 3\operatorname{sen}(x) \cdot f'(x) + f'(x^2) \cdot 2x + f'(\operatorname{sen}(x)) \cdot \cos(x)$$

$$g'(0) = 3\underbrace{\cos(0)}_{=1} \cdot \underbrace{f(0)}_{=-2} + 3\underbrace{\operatorname{sen}(0)}_{=0} \cdot f'(0) + f'(0) \cdot 0 + \underbrace{f'(0)}_{=9} \cdot \underbrace{\cos(0)}_{=1}$$

$$g'(0) = 3 \cdot (-2) + 9 = -6 + 9 = \boxed{3}$$

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